2.1 Measurable Transformations

Def 2.1.1: For a nonempty set Ω and a σ-algebra $\mathcal{F}$ on Ω, $(\Omega, \mathcal{F})$ is a measurable space. If μ is a measure on $(\Omega, \mathcal{F})$, then $(\Omega, \mathcal{F}, \mu)$ is a measure space; it is a probability space if $\mu(\Omega) = 1$.

Def 2.1.2: $f : \Omega \to \mathbb{R}$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R})\rangle$-measurable (or $\mathcal{F}$-measurable) if for each $a \in \mathbb{R}$

$$f^{-1}\big((-\infty, a]\big) \equiv \{\omega : f(\omega) \le a\} \in \mathcal{F}.$$

A random variable is a real valued $\mathcal{F}$-measurable function on a probability space $(\Omega, \mathcal{F}, P)$.

Def 2.1.3: Let $(\Omega_i, \mathcal{F}_i)$, $i = 1, 2$ be measurable spaces. $T : \Omega_1 \to \Omega_2$ is $\langle \mathcal{F}_1, \mathcal{F}_2\rangle$-measurable if

$$T^{-1}(A) \in \mathcal{F}_1 \quad \text{for all } A \in \mathcal{F}_2.$$

- T stays measurable for any $\tilde{\mathcal{F}}_1 \supset \mathcal{F}_1$ and any $\tilde{\mathcal{F}}_2 \subset \mathcal{F}_2$

Prop 2.1.1: Let $(\Omega_i, \mathcal{F}_i)$, $i = 1, 2, 3$ be measurable spaces.

- (Generators) If $\mathcal{F}_2 = \sigma\langle \mathcal{A}\rangle$ and $T^{-1}(A) \in \mathcal{F}_1$ for all $A \in \mathcal{A}$, then T is $\langle \mathcal{F}_1, \mathcal{F}_2\rangle$-measurable
- (Composition) If T_1 is $\langle \mathcal{F}_1, \mathcal{F}_2\rangle$-measurable and T_2 is $\langle \mathcal{F}_2, \mathcal{F}_3\rangle$-measurable, then $T_2 \circ T_1$ is $\langle \mathcal{F}_1, \mathcal{F}_3\rangle$-measurable

Prop 2.1.2: If $f : \mathbb{R}^p \to \mathbb{R}^k$ is continuous, then f is $\langle \mathcal{B}(\mathbb{R}^p), \mathcal{B}(\mathbb{R}^k)\rangle$-measurable.

Prop 2.1.3: Let $f_1, \ldots, f_k$ be $\langle \mathcal{F}, \mathcal{B}(\mathbb{R})\rangle$-measurable from Ω to $\mathbb{R}$. Then

- $f = (f_1, \ldots, f_k)$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}^k)\rangle$-measurable
- $\sum_{i=1}^k f_i$ and $\prod_{i=1}^k f_i$ are $\langle \mathcal{F}, \mathcal{B}(\mathbb{R})\rangle$-measurable
- $\psi \circ f$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}^p)\rangle$-measurable for any continuous $\psi : \mathbb{R}^k \to \mathbb{R}^p$

Corollary 2.1.4: The $\langle \mathcal{F}, \mathcal{B}(\mathbb{R})\rangle$-measurable functions are closed under pointwise addition, multiplication and scalar multiplication.

Prop 2.1.5: Let $f_n : \Omega \to \bar{\mathbb{R}}$ be $\langle \mathcal{F}, \mathcal{B}(\bar{\mathbb{R}})\rangle$-measurable, $n \ge 1$, where $\mathcal{B}(\bar{\mathbb{R}}) \equiv \sigma\langle \mathcal{B}(\mathbb{R}) \cup \{+\infty\} \cup \{-\infty\}\rangle$. Then

- $\sup_n f_n$, $\inf_n f_n$, $\limsup_n f_n$, $\liminf_n f_n$ are $\langle \mathcal{F}, \mathcal{B}(\bar{\mathbb{R}})\rangle$-measurable
- $A \equiv \{\omega : \lim_n f_n(\omega) \text{ exists and is finite}\} \in \mathcal{F}$ and $h \equiv (\lim_n f_n) \cdot I_A$ is $\langle \mathcal{F}, \mathcal{B}(\bar{\mathbb{R}})\rangle$-measurable

Def 2.1.4: For a family $\{f_\lambda : \lambda \in \Lambda\}$ of maps from Ω_1 into Ω_2 and a σ-algebra $\mathcal{F}_2$ on Ω_2,

$$\sigma\langle \{f_\lambda : \lambda \in \Lambda\}\rangle \equiv \sigma\langle \{f_\lambda^{-1}(A) : A \in \mathcal{F}_2, \lambda \in \Lambda\}\rangle$$

is the σ-algebra generated by $\{f_\lambda\}$: the smallest σ-algebra on Ω_1 making every f_λ measurable.

- $\sigma\langle \{I_A\}\rangle = \{\Omega_1, \emptyset, A, A^c\}$
- Coordinate maps $f_i(x) = x_i$ on $\mathbb{R}^k$: $\sigma\langle \{f_i : 1 \le i \le k\}\rangle = \mathcal{B}(\mathbb{R}^k)$

Prop 2.1.6: If Λ is uncountable, then for any $B \in \sigma\langle \{f_\lambda : \lambda \in \Lambda\}\rangle$ there is a countable $\Lambda_B \subset \Lambda$ with $B \in \sigma\langle \{f_\lambda : \lambda \in \Lambda_B\}\rangle$.

2.2 Induced Measures, Distribution Functions

Prop 2.2.1 / Def 2.2.1: Let $T : \Omega_1 \to \Omega_2$ be $\langle \mathcal{F}_1, \mathcal{F}_2\rangle$-measurable and μ a measure on $(\Omega_1, \mathcal{F}_1)$. Then

$$\mu T^{-1}(A) \equiv \mu\big(T^{-1}(A)\big), \quad A \in \mathcal{F}_2$$

is a measure on $\mathcal{F}_2$, called the measure induced by T.

Def 2.2.2: The probability distribution (law) of a random variable X on $(\Omega, \mathcal{F}, P)$ is the induced measure $P_X(A) \equiv P(X^{-1}(A))$, $A \in \mathcal{B}(\mathbb{R})$.

Def 2.2.3: The cdf of X is $F_X(x) \equiv P_X((-\infty, x])$, $x \in \mathbb{R}$.

Prop 2.2.2: Let F be the cdf of a random variable X. Then

- $x_1 < x_2 \Rightarrow F(x_1) \le F(x_2)$ (nondecreasing)
- $F(x) = \lim_{y \downarrow x} F(y)$ (right continuous)
- $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to +\infty} F(x) = 1$

Def 2.2.4: Any $F : \mathbb{R} \to \mathbb{R}$ satisfying the three properties of Prop 2.2.2 is called a cdf.

- Conversely, given a cdf F, $X(x) \equiv x$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_F)$ has $P_X = \mu_F$ and $F_X = F$
- $P(X < x) = F_X(x-) \equiv \lim_{y \uparrow x} F_X(y)$ and $P(X = x) = F_X(x) - F_X(x-)$
- $\{a \in \mathbb{R} : P(X = a) > 0\}$ is countable

Def 2.2.5: X is discrete if there is a countable $A \subset \mathbb{R}$ with $P(X \in A) = 1$. X is continuous if $P(X = x) = 0$ for all $x \in \mathbb{R}$ (iff F_X is continuous on $\mathbb{R}$).

- Any cdf: $F = \alpha F_d + (1 - \alpha) F_c$, with F_d discrete, F_c continuous, $\alpha = \sum_{y \in A} p(y)$, $p(x) = F(x) - F(x-)$, $A = \{x : p(x) > 0\}$

Prop 2.2.3: Let F be the cdf of $X = (X_1, X_2)$, i.e. $F(x) = P(X \le x)$. Then

- For $(x_1, x_2) \le (y_1, y_2)$, $F(y_1, y_2) - F(x_1, y_2) - F(y_1, x_2) + F(x_1, x_2) \ge 0$
- F is right continuous on $\mathbb{R}^2$
- $\lim_{x_1 \to -\infty} F(x_1, a) = \lim_{x_2 \to -\infty} F(a, x_2) = 0$ for all a; $\lim_{x_1, x_2 \to \infty} F(x_1, x_2) = 1$
- As $y \uparrow \infty$: $F(a, y) \to P(X_1 \le a)$ and $F(y, a) \to P(X_2 \le a)$

2.3 Integration

Def 2.3.1: $f : \Omega \to \bar{\mathbb{R}}$ is simple if for some distinct $c_1, \ldots, c_k \in \bar{\mathbb{R}}$ and $A_1, \ldots, A_k \in \mathcal{F}$, $k \in \mathbb{N}$,

$$f = \sum_{i=1}^k c_i I_{A_i}.$$

Def 2.3.2: For a simple nonnegative f on $(\Omega, \mathcal{F}, \mu)$,

$$\int f \, d\mu \equiv \sum_{i=1}^k c_i \mu(A_i), \qquad \text{with } 0 \cdot \infty = 0.$$

The value does not depend on the representation of f.

Prop 2.3.1: Let f, g be simple nonnegative functions on $(\Omega, \mathcal{F}, \mu)$. Then

- (Linearity) For $\alpha, \beta \ge 0$, $\int (\alpha f + \beta g) d\mu = \alpha \int f \, d\mu + \beta \int g \, d\mu$
- (Monotonicity) $f \ge g$ a.e. $(\mu) \Rightarrow \int f \, d\mu \ge \int g \, d\mu$
- $f = g$ a.e. $(\mu) \Rightarrow \int f \, d\mu = \int g \, d\mu$

Def 2.3.3: For a nonnegative measurable $f : \Omega \to$

$\bar{\mathbb{R}}_+$,

$$\int f\,d\mu \equiv \lim_{n\to\infty}\int f_n\,d\mu$$

where $\{f_n\}_{n\geq 1}$ is any sequence of nonnegative simple functions with $f_n(\omega)\uparrow f(\omega)$ for all ω.

Prop 2.3.2: If $\{f_n\}$, $\{g_n\}$ are simple nonnegative with $f_n\uparrow f$ and $g_n\uparrow f$, then $\lim_n\int f_n\,d\mu = \lim_n\int g_n\,d\mu$. (Also valid if the convergence is only a.e. (μ).)

- Standard choice: $f_n = j2^{-n}$ on $\{j2^{-n}\leq f < (j+1)2^{-n}\}$, $j=0,\ldots,n2^n-1$, and $f_n=n$ on $\{f\geq n\}$
- $\int f\,d\mu = \sup\{\int g\,d\mu : g \text{ simple}, 0\leq g\leq f\}$
- $\Omega=\{\omega_i\}_{i\geq 1}$, $\mathcal{F}=\mathcal{P}(\Omega)$: $\int f\,d\mu = \sum_{i=1}^\infty f(\omega_i)\mu(\{\omega_i\})$

Corollary 2.3.3: Prop 2.3.1 remains valid for nonnegative measurable f and g.

Theorem 2.3.4 (MCT): Let $\{f_n\}_{n\geq 1}$ and f be nonnegative measurable functions with $f_n\uparrow f$ a.e. (μ). Then

$$\int f\,d\mu = \lim_{n\to\infty}\int f_n\,d\mu.$$

Corollary 2.3.5: For nonnegative measurable $\{h_n\}_{n\geq 1}$,

$$\int\Big(\sum_{n=1}^\infty h_n\Big)d\mu = \sum_{n=1}^\infty\int h_n\,d\mu.$$

Corollary 2.3.6: For nonnegative measurable f, $\nu(A)\equiv\int fI_A\,d\mu$, $A\in\mathcal{F}$, is a measure on $(\Omega,\mathcal{F})$.

- $\mu(A)=0\Rightarrow\nu(A)=0$, i.e. ν is dominated by μ

Theorem 2.3.7 (Fatou's lemma): For nonnegative measurable $\{f_n\}_{n\geq 1}$,

$$\int\liminf_{n\to\infty} f_n\,d\mu \leq \liminf_{n\to\infty}\int f_n\,d\mu.$$

- Can be strict: $f_n = I_{[n,\infty)}$ on $(\mathbb{R},\mathcal{B}(\mathbb{R}),m)$; $f_n = nI_{[0,1/n]}$ on $([0,1],\mathcal{B}([0,1]),m)$

Def 2.3.4: For measurable f, let $f^+ = fI_{\{f\geq 0\}}$ and $f^- = -fI_{\{f<0\}}$, so $f=f^+-f^-$, $|f|=f^++f^-$. Then

$$\int f\,d\mu \equiv \int f^+d\mu - \int f^-d\mu$$

provided at least one integral on the right is finite.

Def 2.3.5: f is integrable w.r.t. μ if $\int|f|d\mu<\infty$ (iff $\int f^+d\mu<\infty$ and $\int f^-d\mu<\infty$).

Def 2.3.6: For $A\in\mathcal{F}$, $\int_A f\,d\mu\equiv\int fI_A\,d\mu$.

Def 2.3.7 (L^p-spaces): For $0<p<\infty$,

$$L^p(\Omega,\mathcal{F},\mu)\equiv\Big\{f:\int|f|^pd\mu<\infty\Big\},$$

and $L^\infty(\Omega,\mathcal{F},\mu)\equiv\{f:\mu(\{|f|>K\})=0$ for some $K\in(0,\infty)\}$.

Prop 2.3.8: Let $f,g\in L^1(\Omega,\mathcal{F},\mu)$. Then

- $\int(\alpha f+\beta g)d\mu=\alpha\int f\,d\mu+\beta\int g\,d\mu$ for $\alpha,\beta\in\mathbb{R}$
- $f\geq g$ a.e. $(\mu)\Rightarrow\int f\,d\mu\geq\int g\,d\mu$
- $f=g$ a.e. $(\mu)\Rightarrow\int f\,d\mu=\int g\,d\mu$
- L^1 is a vector space; $\|f\|_1\equiv\int|f|d\mu$ is a norm once $f\sim g$ iff $f=g$ a.e. (μ)

Prop 2.3.9: If $f\geq 0$ a.e. (μ), then $\int f\,d\mu=0$ iff $f=0$ a.e. (μ).

Prop 2.3.10: $f\in L^1(\Omega,\mathcal{F},\mu)\Rightarrow|f|<\infty$ a.e. (μ).

Theorem 2.3.11 (EDCT): Let f_n,g_n be measurable with $|f_n|\leq g_n$ a.e. (μ) for all $n\geq 1$. Suppose

- $g_n\to g$ a.e. (μ) and $f_n\to f$ a.e. (μ)
- $g_n,g\in L^1(\Omega,\mathcal{F},\mu)$ and $\int|g_n|d\mu\to\int|g|d\mu$

Then $f\in L^1(\Omega,\mathcal{F},\mu)$,

$$\lim_{n\to\infty}\int f_n\,d\mu=\int f\,d\mu \text{ and } \lim_{n\to\infty}\int|f_n-f|d\mu=0.$$

Corollary 2.3.12 (DCT): If $|f_n|\leq g$ a.e. (μ) for all $n\geq 1$, $\int g\,d\mu<\infty$ and $f_n\to f$ a.e. (μ), then $f\in L^1(\Omega,\mathcal{F},\mu)$ and the conclusions of the EDCT hold.

Corollary 2.3.13 (BCT): Let $\mu(\Omega)<\infty$. If $|f_n|\leq k$ a.e. (μ) for some $0<k<\infty$ and $f_n\to f$ a.e. (μ), then the conclusions of the EDCT hold.

Theorem 2.3.14 (Approximation): Let μ_F be a Lebesgue-Stieltjes measure on $(\mathbb{R},\mathcal{B}(\mathbb{R}))$ and $f\in L^p(\mathbb{R},\mathcal{B}(\mathbb{R}),\mu_F)$, $0<p<\infty$. For any $\delta>0$ there exist a step function h and a continuous g with compact support such that

$$\int|f-h|^pd\mu_F<\delta,\qquad\int|f-g|^pd\mu_F<\delta.$$

- Step function: $h=\sum_{i=1}^k c_iI_{A_i}$, A_i bounded disjoint intervals
- Fails for $p=\infty$ (e.g. $f\equiv 1$ in $L^\infty(m)$)

2.4 Riemann and Lebesgue Integrals

Let f be bounded on $[a,b]$, $P=\{a=x_0<x_1<\cdots<x_n=b\}$, $M_i=\sup\{f(x):x_i\leq x\leq x_{i+1}\}$, $m_i=\inf\{f(x):x_i\leq x\leq x_{i+1}\}$.

Def 2.4.1: Upper and lower Riemann sums:

$$U(f,P)\equiv\textstyle\sum_{i=0}^{n-1}M_i(x_{i+1}-x_i),$$
$$L(f,P)\equiv\textstyle\sum_{i=0}^{n-1}m_i(x_{i+1}-x_i).$$

- $P\subset Q\Rightarrow U(f,P)\geq U(f,Q)\geq L(f,Q)\geq L(f,P)$

Def 2.4.2: Upper and lower Riemann integrals:

$$\overline{\int}f=\inf_P U(f,P),\qquad\underline{\int}f=\sup_P L(f,P).$$

Def 2.4.3: f is Riemann integrable if $\overline{\int}f=\underline{\int}f$.

Theorem 2.4.1: Let f be bounded on a bounded interval $[a,b]$. Then f is Riemann integrable on $[a,b]$ iff f is continuous a.e. (m) on $[a,b]$. In this case f is Lebesgue integrable on $[a,b]$ and $\int_{[a,b]}f\,dm$ equals the Riemann integral.

- Lebesgue integrable $\not\Rightarrow$ Riemann integrable: $f=I_{\mathbb{Q}\cap[0,1]}$
- Riemann partitions the domain of f; Lebesgue partitions the range

2.5 More on Convergence

Let $\{f_n\}_{n\geq 1}$, f be measurable from $(\Omega,\mathcal{F},\mu)$ to $\bar{\mathbb{R}}$.

Def 2.5.1 (pointwise): $\lim_{n\to\infty}f_n(\omega)=f(\omega)$ for all $\omega\in\Omega$.

Def 2.5.2 (a.e.): $f_n \to f$ a.e. (μ) if there is $B \in \mathcal{F}$ with $\mu(B) = 0$ and $\lim_{n\to\infty} f_n(\omega) = f(\omega)$ for all $\omega \in B^c$.

Def 2.5.3 (in measure): $f_n \xrightarrow{m} f$ if for each $\epsilon > 0$,

$$\lim_{n\to\infty} \mu\big(\{|f_n - f| > \epsilon\}\big) = 0.$$

Def 2.5.4 (in L^p): For $0 < p < \infty$, $f_n \xrightarrow{L^p} f$ if $\int |f_n|^p d\mu < \infty$ for all n, $\int |f|^p d\mu < \infty$ and

$$\lim_{n\to\infty} \int |f_n - f|^p d\mu = 0,$$

equivalently $\|f_n - f\|_p \to 0$, where

$$\|g\|_p = \Big(\int |g|^p d\mu\Big)^{\min\{1/p,\,1\}}.$$

Def 2.5.5 (uniform): $\lim_{n\to\infty} \sup\{|f_n(\omega) - f(\omega)| : \omega \in \Omega\} = 0$.

Def 2.5.6 (in L^∞): $\lim_{n\to\infty} \|f_n - f\|_\infty = 0$, where

$$\|g\|_\infty = \inf\{K \in (0,\infty) : \mu(\{|g| > K\}) = 0\}.$$

Def 2.5.7 (nearly uniform): For every $\epsilon > 0$ there is $A \in \mathcal{F}$ with $\mu(A) < \epsilon$ and $\sup\{|f_n(\omega) - f(\omega)| : \omega \in A^c\} \to 0$.

- $f_n = nI_{[0,1/n]}$ on $([0,1], \mathcal{B}([0,1]), m)$: $f_n \to 0$ nearly uniformly but not in $L^\infty(m)$

Theorem 2.5.1: If $\mu(\Omega) < \infty$, then $f_n \to f$ a.e. (μ) implies $f_n \xrightarrow{m} f$.

- Fails if $\mu(\Omega) = \infty$: $f_n = I_{[n,\infty)}$ on $\mathbb{R}$ with m

Theorem 2.5.2: If $f_n \xrightarrow{m} f$, then there is a subsequence $\{n_k\}_{k\ge1}$ such that $f_{n_k} \to f$ a.e. (μ).

- The EDCT remains valid with a.e. convergence of $\{f_n\}$, $\{g_n\}$ replaced by convergence in measure

Theorem 2.5.3: If $f_n \xrightarrow{L^p} f$ for some $0 < p < \infty$, then $f_n \xrightarrow{m} f$.

- Converse fails: $f_n = nI_{[0,1/n]}$, $f \equiv 0$ on $[0,1]$; $f_n \xrightarrow{m} f$ but $\int |f_n - f| dm \equiv 1$

Theorem 2.5.4 (Scheffé): Let f_n, f be nonnegative measurable with $f_n \to f$ a.e. (μ), $\int f_n\, d\mu \to \int f\, d\mu$ and $\int f\, d\mu < \infty$. Then

$$\lim_{n\to\infty} \int |f_n - f| d\mu = 0.$$

Corollary 2.5.5: If f_n, f are probability densities w.r.t. μ ($\int f_n\, d\mu = \int f\, d\mu = 1$, $f_n, f \ge 0$ a.e. (μ)) and $f_n \to f$ a.e. (μ), then $\int |f_n - f| d\mu \to 0$.

- Both remain valid with $f_n \xrightarrow{m} f$ in place of a.e. convergence

Corollary 2.5.6: If $p_{nk}, p_k \ge 0$, $\sum_k p_{nk} = 1 = \sum_k p_k$ and $p_{nk} \to p_k$ for each k, then $\sum_{k=1}^\infty |p_{nk} - p_k| \to 0$.

Def 2.5.8 (UI): $\{f_\lambda : \lambda \in \Lambda\} \subset L^1(\Omega, \mathcal{F}, \mu)$ is uniformly integrable if

$$\sup_{\lambda\in\Lambda} a_\lambda(t) \to 0 \ \text{ as } t \to \infty, \quad a_\lambda(t) \equiv \int_{\{|f_\lambda| > t\}} |f_\lambda| d\mu.$$

Prop 2.5.7: Let $\{f_\lambda : \lambda \in \Lambda\}$ be μ-integrable.

- Λ finite $\Rightarrow$ UI
- $\sup_\lambda \int |f_\lambda|^{1+\epsilon} d\mu < \infty$ for some $\epsilon > 0 \Rightarrow$ UI
- $|f_\lambda| \le g$ a.e. (μ) and $\int g\, d\mu < \infty \Rightarrow$ UI
- $\{f_\lambda\}$ and $\{g_\gamma\}$ UI $\Rightarrow \{f_\lambda + g_\gamma : \lambda \in \Lambda, \gamma \in \Gamma\}$ UI
- UI and $\mu(\Omega) < \infty \Rightarrow \sup_\lambda \int |f_\lambda| d\mu < \infty$
- Converse of the last fails: $f_n = nI_{[0,1/n]}$ on $[0,1]$
- (Rmk 2.5.3) If $\phi : \mathbb{R}_+ \to \mathbb{R}_+$ is nondecreasing with $\phi(x)/x \uparrow \infty$ and $\sup_\lambda \int \phi(|f_\lambda|) d\mu < \infty$, then UI; e.g. x^k $(k > 1)$, $x(\log x)^\beta I(x > 1)$, $e^{\beta x}$ $(\beta > 0)$

Prop 2.5.8: Let $f \in L^1(\Omega, \mathcal{F}, \mu)$. For every $\epsilon > 0$ there is $\delta > 0$ such that $\mu(A) < \delta \Rightarrow \int_A |f| d\mu < \epsilon$.

Def 2.5.9 (u.a.c.): With $\nu_{|f_\lambda|}(A) \equiv \int_A |f_\lambda| d\mu$, the measures $\{\nu_{|f_\lambda|} : \lambda \in \Lambda\}$ are uniformly absolutely continuous w.r.t. μ if for every $\epsilon > 0$ there is $\delta > 0$ with

$$\mu(A) < \delta \Rightarrow \sup\{\nu_{|f_\lambda|}(A) : \lambda \in \Lambda\} < \epsilon.$$

Theorem 2.5.9: Let $\{f_\lambda\} \subset L^1(\Omega, \mathcal{F}, \mu)$ and $\mu(\Omega) < \infty$. Then $\{f_\lambda\}$ is UI iff $\sup_\lambda \int |f_\lambda| d\mu < \infty$ and $\{\nu_{|f_\lambda|}\}$ is u.a.c. (μ).

Theorem 2.5.10: Let $\mu(\Omega) < \infty$ and $\{f_n\} \subset L^1(\Omega, \mathcal{F}, \mu)$ with $f_n \to f$ a.e. (μ), f measurable. If $\{f_n\}$ is UI, then f is integrable and

$$\lim_{n\to\infty} \int |f_n - f| d\mu = 0.$$

- UI is sufficient, not necessary, for $\int f_n\, d\mu \to \int f\, d\mu$
- If $f_n \ge 0$ and $\mu(\Omega) < \infty$: $\int f_n\, d\mu \to \int f\, d\mu < \infty$ iff $\{f_n\}$ UI
- If $\mu(\Omega) = \infty$, UI is no longer sufficient

Theorem 2.5.11 (Egorov): If $f_n \to f$ a.e. (μ) and $\mu(\Omega) < \infty$, then $f_n \to f$ nearly uniformly (μ).

Theorem 2.5.12 (Lusin): Let $F : \mathbb{R} \to \mathbb{R}$ be nondecreasing, $\mu = \mu_F^*$ on $(\mathbb{R}, \mathcal{M}_{\mu_F^*})$, and $f : \mathbb{R} \to \bar{\mathbb{R}}$ be $\langle \mathcal{M}_{\mu_F^*}, \mathcal{B}(\bar{\mathbb{R}})\rangle$-measurable with $\mu(\{x : |f(x)| = \infty\}) = 0$. Then for every $\epsilon > 0$ there is a continuous $g : \mathbb{R} \to \mathbb{R}$ with $\mu(\{x : f(x) \ne g(x)\}) < \epsilon$.

Summary of modes:

- a.e. $\Rightarrow$ in measure (if $\mu(\Omega) < \infty$)
- a.e. $\Rightarrow$ nearly uniform (if $\mu(\Omega) < \infty$)
- in measure $\Rightarrow$ a.e. along a subsequence
- $L^p \Rightarrow$ in measure $(0 < p < \infty)$
- a.e. (or in measure) + domination $\Rightarrow$ L^1 (DCT)
- a.e. + UI + $\mu(\Omega) < \infty \Rightarrow L^1$
- a.e. + $f_n, f \ge 0$ + $\int f_n \to \int f < \infty \Rightarrow L^1$ (Scheffé)

3.1 Inequalities

Theorem 3.1.1 (Markov): For nonnegative measurable f on $(\Omega, \mathcal{F}, \mu)$ and any $0 < t < \infty$,

$$\mu(\{f \ge t\}) \le \frac{\int f\, d\mu}{t}.$$

Corollary 3.1.2: For a random variable X, $r > 0$, $t > 0$,

$$P(|X| \ge t) \le \frac{E|X|^r}{t^r}.$$

Corollary 3.1.3 (Chebychev): If $EX^2 < \infty$, $E(X) = \mu$, $\mathrm{Var}(X) = \sigma^2$, then for any $0 < k < \infty$,

$$P(|X - \mu| \ge k\sigma) \le \frac{1}{k^2}.$$

Corollary 3.1.4: For $\phi : \mathbb{R}_+ \to \mathbb{R}_+$ nondecreasing and $0 < t < \infty$,

$$P(|X| \geq t) \leq \frac{E\phi(|X|)}{\phi(t)}.$$

Corollary 3.1.5 (Cramér): For any random variable X and $t > 0$,

$$P(X \geq t) \leq \inf_{\theta > 0} \frac{E(e^{\theta X})}{e^{\theta t}}.$$

Def 3.1.1: $\phi : (a, b) \to \mathbb{R}$ is convex if for all $0 \leq \lambda \leq 1$, $a < x \leq y < b$,

$$\phi(\lambda x + (1-\lambda)y) \leq \lambda\phi(x) + (1-\lambda)\phi(y).$$

Prop 3.1.6: ϕ is convex iff for all $a < x_1 < x_2 < x_3 < b$,

$$\frac{\phi(x_2) - \phi(x_1)}{x_2 - x_1} \leq \frac{\phi(x_3) - \phi(x_1)}{x_3 - x_1} \leq \frac{\phi(x_3) - \phi(x_2)}{x_3 - x_2}.$$

Prop 3.1.7: Let $\phi : (a, b) \to \mathbb{R}$ be convex. Then

- $\phi'_+(x) \equiv \lim_{y \downarrow x} \frac{\phi(y)-\phi(x)}{y-x}$ and $\phi'_-(x) \equiv \lim_{y \uparrow x} \frac{\phi(y)-\phi(x)}{y-x}$ exist and are finite for each $x \in (a, b)$
- $\phi'_- \leq \phi'_+$ and both are nondecreasing on (a, b)
- ϕ' exists except on the countable set of discontinuity points of ϕ'_+ and ϕ'_-
- ϕ is Lipschitz on $[c, d]$ for any $a < c < d < b$: $|\phi(x) - \phi(y)| \leq K|x - y|$
- (Supporting line) For any $a < c, x < b$, $\phi(x) - \phi(c) \geq \phi'_+(c)(x-c)$ and $\phi(x) - \phi(c) \geq \phi'_-(c)(x-c)$

Prop 3.1.8: If ϕ is twice differentiable on (a, b) and $\phi'' \geq 0$ on (a, b), then ϕ is convex on (a, b).

- Convex: $|x|^p$ $(p \geq 1)$ and e^x on $\mathbb{R}$; $-\log x$ and $x \log x$ on $(0, \infty)$

Theorem 3.1.9 (Jensen): Let f be measurable on a probability space $(\Omega, \mathcal{F}, P)$ with $P(f \in (a, b)) = 1$, $-\infty \leq a < b \leq \infty$, and let $\phi : (a, b) \to \mathbb{R}$ be convex. Then

$$\phi\Big(\int f\, dP\Big) \leq \int \phi(f)\, dP,$$

provided $\int |f| dP < \infty$ and $\int |\phi(f)| dP < \infty$.

- Random variables: $\phi(EX) \leq E\phi(X)$
- If ϕ is strictly convex, equality holds iff $f = \int f\, dP$ a.e. (P)

Prop 3.1.10: Let $k \geq 1$, $p_1, \ldots, p_k > 0$ with $\sum_{i=1}^k p_i = 1$.

- For real a_i: $\sum_{i=1}^k p_i \exp(a_i) \geq \exp\big(\sum_{i=1}^k p_i a_i\big)$
- For $b_i \geq 0$: $\sum_{i=1}^k p_i b_i \geq \prod_{i=1}^k b_i^{p_i}$, with equality iff $b_1 = \cdots = b_k$
- (AM–GM) $\frac{1}{k}\sum_{i=1}^k b_i \geq \big(\prod_{i=1}^k b_i\big)^{1/k}$
- For a, b real and $1 \leq p < \infty$: $|a + b|^p \leq 2^{p-1}(|a|^p + |b|^p)$

Theorem 3.1.11 (Hölder): Let $1 < p < \infty$, $q = \frac{p}{p-1}$, $f \in L^p(\Omega, \mathcal{F}, \mu)$ and $g \in L^q(\Omega, \mathcal{F}, \mu)$. Then

$$\int |fg| d\mu \leq \Big(\int |f|^p d\mu\Big)^{\frac{1}{p}} \Big(\int |g|^q d\mu\Big)^{\frac{1}{q}},$$

i.e. $\|fg\|_1 \leq \|f\|_p \|g\|_q$.

- If $\|fg\|_1 \neq 0$, equality holds iff $|f|^p = c|g|^q$ a.e. (μ) for some $c \in (0, \infty)$
- $(p = 1,\ q = \infty)$ $f \in L^1$, $g \in L^\infty \Rightarrow \|fg\|_1 \leq \|f\|_1 \|g\|_\infty$; equality $\Rightarrow |g| = \|g\|_\infty$ on $\{|f| > 0\}$ a.e. (μ)

Corollary 3.1.12 (Cauchy-Schwarz): For $f, g \in L^2(\Omega, \mathcal{F}, \mu)$,

$$\int |fg| d\mu \leq \Big(\int |f|^2 d\mu\Big)^{\frac{1}{2}} \Big(\int |g|^2 d\mu\Big)^{\frac{1}{2}}.$$

Corollary 3.1.13: For real a_i, b_i, positive c_i, $1 < p < \infty$, $q = \frac{p}{p-1}$,

$$\sum_{i=1}^k |a_i b_i| c_i \leq \Big(\sum_{i=1}^k |a_i|^p c_i\Big)^{\frac{1}{p}} \Big(\sum_{i=1}^k |b_i|^q c_i\Big)^{\frac{1}{q}}.$$

Theorem 3.1.14 (Minkowski): Let $1 < p < \infty$ and $f, g \in L^p(\Omega, \mathcal{F}, \mu)$. Then

$$\Big(\int |f + g|^p d\mu\Big)^{\frac{1}{p}} \leq \Big(\int |f|^p d\mu\Big)^{\frac{1}{p}} + \Big(\int |g|^p d\mu\Big)^{\frac{1}{p}},$$

i.e. $\|f + g\|_p \leq \|f\|_p + \|g\|_p$. Also holds for $p = 1$ and $p = \infty$.

3.2 L^p-Spaces

Write $L^p(\mu)$ for $L^p(\Omega, \mathcal{F}, \mu)$, with $\|f\|_p$ as in Def 2.5.4 and $\|f\|_\infty \equiv \sup\{k : \mu(\{|f| > k\}) > 0\}$ (the essential supremum).

Prop 3.2.1: For $0 < p \leq \infty$, $f, g \in L^p(\mu)$ and $a, b \in \mathbb{R} \Rightarrow af + bg \in L^p(\mu)$.

- For $0 < p \leq 1$ and $x, y \geq 0$: $(x + y)^p \leq x^p + y^p$

Def 3.2.1: For $f, g \in L^p(\mu)$, $0 < p \leq \infty$, $d_p(f, g) \equiv \|f - g\|_p$.

Def 3.2.2: $f \sim g$ (f is equivalent to g) if $f = g$ a.e. (μ).

- Identifying f with its class $[f]$, $(L^p(\mu), d_p)$ is a metric space for $1 \leq p \leq \infty$
- For $0 < p < 1$, $d_p(f, g) \equiv \int |f - g|^p d\mu$ is a metric

Theorem 3.2.2: For $0 < p \leq \infty$, $(L^p(\mu), d_p)$ is complete: every Cauchy sequence converges to an element of $L^p(\mu)$.

Def 3.2.3: $T : L^p(\mu) \to \mathbb{R}$ is a linear functional if $T(\alpha_1 f_1 + \alpha_2 f_2) = \alpha_1 T(f_1) + \alpha_2 T(f_2)$. It is bounded if $|T(f)| \leq c\|f\|_p$ for all f, some $c \in (0, \infty)$, with norm

$$\|T\| = \sup\{|Tf| : f \in L^p(\mu), \|f\|_p = 1\}.$$

Def 3.2.4: The dual space $(L^p(\mu))^*$ is the set of all continuous linear functionals on $L^p(\mu)$.

Theorem 3.2.3 (Riesz representation): Let $1 \leq p < \infty$ and $T : L^p(\mu) \to \mathbb{R}$ be linear and continuous. Then there is $g \in L^q(\mu)$ such that

$$T(f) = T_g(f) \equiv \int fg\, d\mu \quad \text{for all } f \in L^p(\mu),$$

where $q = \frac{p}{p-1}$ for $1 < p < \infty$ and $q = \infty$ if $p = 1$.

- $|T_g(f)| \leq \|g\|_q \|f\|_p$ and $\|T_g\| = \|g\|_q$
- Not valid for $p = \infty$

4.1 The Lebesgue-Radon-Nikodym Theorem

Def 4.1.1: Let μ, ν be measures on $(\Omega, \mathcal{F})$. μ is

dominated by (absolutely continuous w.r.t.) ν, written $\mu \ll \nu$, if

$$\nu(A) = 0 \Rightarrow \mu(A) = 0 \quad \text{for all } A \in \mathcal{F}.$$

- $\mu(A) \equiv \int_A f\, d\nu$ with $f \geq 0$ measurable $\Rightarrow \mu \ll \nu$

Def 4.1.2: μ is singular w.r.t. ν, written $\mu \perp \nu$, if there is $B \in \mathcal{F}$ such that

$$\mu(B) = 0 \quad \text{and} \quad \nu(B^c) = 0.$$

- Symmetric: $\mu \perp \nu$ iff $\nu \perp \mu$ ($\ll$ is not symmetric)
- Then $\mu(A) = \mu(A \cap B^c)$ and $\nu(A) = \nu(A \cap B)$ for all $A \in \mathcal{F}$
- E.g. Lebesgue measure and counting measure on $\mathbb{Z}$ ($B = \mathbb{Z}$)

Theorem 4.1.1: Let μ_1, μ_2 be σ-finite measures on $(\Omega, \mathcal{F})$.

- (Lebesgue decomposition) μ_1 can be uniquely decomposed as $\mu_1 = \mu_{1a} + \mu_{1s}$, where μ_{1a}, μ_{1s} are σ-finite measures with $\mu_{1a} \ll \mu_2$ and $\mu_{1s} \perp \mu_2$
- (Radon-Nikodym) There is a nonnegative measurable h on $(\Omega, \mathcal{F})$ such that

$$\mu_{1a}(A) = \int_A h\, d\mu_2 \quad \text{for all } A \in \mathcal{F}$$

- σ-finiteness cannot be dropped: $\mu = m$, $\nu =$ counting measure on $[0,1]$ has $\mu \ll \nu$ but no density
- If ν is σ-finite and $\mu \ll \nu$ (μ not necessarily σ-finite), Radon-Nikodym still holds

Def 4.1.3: If $h \geq 0$ is measurable and $\mu(A) = \int_A h\, d\nu$ for all $A \in \mathcal{F}$, then h is the Radon-Nikodym derivative of μ w.r.t. ν, written $\frac{d\mu}{d\nu} = h$.

- For μ σ-finite, $\frac{d\mu}{d\nu}$ is unique up to equivalence a.e. (ν)
- For measurable $f \geq 0$: $\int f\, d\mu = \int f \frac{d\mu}{d\nu}\, d\nu$

Prop 4.1.2: Let $\nu, \mu, \mu_1, \mu_2, \ldots$ be σ-finite measures on $(\Omega, \mathcal{F})$.

- (Chain rule) $\mu_1 \ll \mu_2$ and $\mu_2 \ll \mu_3 \Rightarrow \mu_1 \ll \mu_3$ and $\frac{d\mu_1}{d\mu_3} = \frac{d\mu_1}{d\mu_2}\frac{d\mu_2}{d\mu_3}$ a.e. (μ_3)
- (Linearity) $\mu_1, \mu_2 \ll \mu_3$, $\alpha, \beta \geq 0 \Rightarrow \alpha\mu_1 + \beta\mu_2 \ll \mu_3$ and $\frac{d(\alpha\mu_1+\beta\mu_2)}{d\mu_3} = \alpha\frac{d\mu_1}{d\mu_3} + \beta\frac{d\mu_2}{d\mu_3}$ a.e. (μ_3)
- (Inverse) $\mu \ll \nu$ and $\frac{d\mu}{d\nu} > 0$ a.e. (ν) $\Rightarrow \nu \ll \mu$ and $\frac{d\nu}{d\mu} = \left(\frac{d\mu}{d\nu}\right)^{-1}$ a.e. (μ)
- Let $\mu = \sum_{n=1}^{\infty} \alpha_n \mu_n$ with $\alpha_n > 0$. Then $\mu \ll \nu$ iff $\mu_n \ll \nu$ for each n, in which case $\frac{d\mu}{d\nu} = \sum_{n=1}^{\infty} \alpha_n \frac{d\mu_n}{d\nu}$ a.e. (ν); and $\mu \perp \nu$ iff $\mu_n \perp \nu$ for all n

4.3 Functions of Bounded Variation

Def 4.3.1: For $f : [a,b] \to \mathbb{R}$ and a partition $Q = \{a = x_0 < x_1 < \cdots < x_n = b\}$, the positive, negative and total variations of f w.r.t. Q are

$$P(f,Q) \equiv \sum_{i=1}^n (f(x_i) - f(x_{i-1}))^+$$
$$N(f,Q) \equiv \sum_{i=1}^n (f(x_i) - f(x_{i-1}))^-$$
$$T(f,Q) \equiv \sum_{i=1}^n |f(x_i) - f(x_{i-1})|$$

- $P(f,Q) + N(f,Q) = T(f,Q)$ and $P(f,Q) - N(f,Q) = f(b) - f(a)$

Def 4.3.2: $P(f,[a,b]) \equiv \sup_Q P(f,Q)$, and likewise $N(f,[a,b])$, $T(f,[a,b])$, with the supremum over all finite partitions Q of $[a,b]$.

Def 4.3.3: f is of bounded variation on $[a,b]$, $f \in BV[a,b]$, if $T(f,[a,b]) < \infty$.

Theorem 4.3.1: Let $f \in BV[a,b]$, $f_1(x) \equiv P(f,[a,x])$ and $f_2(x) \equiv N(f,[a,x])$. Then f_1, f_2 are nondecreasing on $[a,b]$ and

$$f(x) - f(a) = f_1(x) - f_2(x), \quad a \leq x \leq b.$$

- $T(f,[a,b]) = P(f,[a,b]) + N(f,[a,b])$
- $f \in BV[a,b]$ iff f is a difference of two nondecreasing functions

Corollary 4.3.2: $f \in BV[a,b]$ iff there is a finite signed measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $f(x) = \mu([a,x])$, $a \leq x \leq b$.

Remark 4.3.2:

- f Lipschitz on $[a,b] \Rightarrow f \in BV[a,b]$; in particular if f is continuous on $[a,b]$, differentiable on (a,b) with f' bounded
- $I_{\mathbb{Q}} \notin BV[a,b]$; $x^2 \sin\frac{1}{x^2} \notin BV[0,1]$; $x^2 \sin\frac{1}{x} \in BV[0,1]$
- f nondecreasing $\Rightarrow f$ differentiable a.e. (m) and $\int_{[a,b]} f'dm \leq f(b) - f(a)$
- $f \in BV[a,b] \Rightarrow f$ differentiable a.e. (m) and $\int_{[a,b]} |f'|dm \leq T(f,[a,b])$

4.4 Absolutely Continuous Functions on $\mathbb{R}$

Def 4.4.1: $F : \mathbb{R} \to \mathbb{R}$ is absolutely continuous (a.c.) if for all $\epsilon > 0$ there is $\delta > 0$ such that if $I_j = [a_j, b_j]$, $j = 1, \ldots, k$ ($k \in \mathbb{N}$) are disjoint and $\sum_{j=1}^k (b_j - a_j) < \delta$, then

$$\sum_{j=1}^k |F(b_j) - F(a_j)| < \epsilon.$$

- F differentiable with F' bounded $\Rightarrow F$ a.c.
- F a.c. $\Rightarrow F$ uniformly continuous
- F a.c. $\Rightarrow F$ of bounded variation on any finite $[a,b]$

Def 4.4.2: $F : [a,b] \to \mathbb{R}$ is a.c. if $\tilde{F}$ is a.c., where $\tilde{F} = F$ on $[a,b]$, $\tilde{F}(x) = F(a)$ for $x < a$, $\tilde{F}(x) = F(b)$ for $x > b$.

- x^2 is a.c. on any bounded interval but not on $\mathbb{R}$

Theorem 4.4.1 (Fundamental theorem of Lebesgue integral calculus): $F : [a,b] \to \mathbb{R}$ is a.c. iff there is a Lebesgue measurable $f : [a,b] \to \mathbb{R}$, integrable w.r.t. m, such that

$$F(x) = F(a) + \int_{[a,x]} f\, dm \quad \text{for all } a \leq x \leq b.$$

Theorem 4.4.2: If F satisfies the representation in Theorem 4.4.1, then F is differentiable a.e. (m) and $F' = f$ a.e. (m).

Theorem 4.4.3: Let $F : \mathbb{R} \to \mathbb{R}$ be nondecreasing with Lebesgue-Stieltjes measure μ_F. Then F is a.c. on $[a,b]$ for all $-\infty < a < b < \infty$ iff $\mu_F \ll m$.

Def 4.4.3: A Radon measure μ on $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$ ($\mu(A) < \infty$ for bounded Borel A) is differentiable at

x with derivative $(D\mu)(x)$ if for any $\epsilon > 0$ there is $\delta > 0$ such that

$$\left|\frac{\mu(A)}{m(A)} - (D\mu)(x)\right| < \epsilon$$

for every open ball $A \ni x$ with $\text{diam}(A) < \delta$.

Theorem 4.4.4: Let μ be a Radon measure on $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$.

- μ is differentiable a.e. (m), $D\mu$ is Lebesgue measurable and ≥ 0 a.e. (m), and $\int_A D\mu\, dm \leq \mu(A)$ for all bounded Borel A
- Let $\mu_a(A) \equiv \int_A D\mu\, dm$ and $\mu_s \equiv \mu - \mu_a$. Then $\mu_s \perp m$ and $D\mu_s = 0$ a.e. (m)
- $\mu \perp m$ iff $D\mu = 0$ a.e. (m); $\mu \ll m$ iff $\mu(A) = \int_A D\mu\, dm$ for all $A \in \mathcal{B}(\mathbb{R}^k)$
- $\mu(A) = \int_A f\, dm \Rightarrow D\mu(x) = f(x)$ for almost all x (m)

Theorem 4.4.5: Let $f : \mathbb{R}^k \to \mathbb{R}_+$ be integrable w.r.t. m on bounded sets. For almost all x (m), for each $\epsilon > 0$ there is $\delta > 0$ such that

$$\frac{1}{m(A)} \int_A |f - f(x)| dm < \epsilon$$

for all open balls $A \ni x$ with $\text{diam}(A) < \delta$.

Theorem 4.4.6 (Change of variables in $\mathbb{R}^k$): Let $V \subset \mathbb{R}^k$ be open and $T = (T_1, \ldots, T_k)$ have partial derivatives on V with Jacobian $J_T \equiv \det\left(\frac{\partial T_i}{\partial x_j}\right)$ continuous and positive on V. Suppose $W = T(V)$ is bounded and open, T is one-to-one and T^{-1} is continuous. Then

- $T(E)$ is a Borel subset of W for all Borel $E \subset V$
- $\nu(\cdot) \equiv m(T(\cdot))$ is a measure with $\nu \ll m$ and $\frac{d\nu}{dm} = J_T$
- For any $h \in L^1(W, m)$,

$$\int_W h\, dm = \int_V h(T(\cdot)) J_T(\cdot)\, dm$$

- $\lambda(\cdot) \equiv mT^{-1}(\cdot)$ is a measure on $\mathcal{B}(W)$ with $\lambda \ll m$ and $\frac{d\lambda}{dm} = |J_T(T^{-1}(\cdot))|^{-1}$
- For $\mu \ll m$ on $\mathcal{B}(V)$, $\psi \equiv \mu T^{-1} \ll m$ with $\frac{d\psi}{dm} = \frac{d\mu}{dm}\big(T^{-1}(\cdot)\big)\big(J_T(T^{-1}(\cdot))\big)^{-1}$ on W

4.5 Singular Distributions

4.5.1 Decomposition of a cdf

Def 4.5.1: A cdf F is singular if $F' \equiv 0$ a.e. w.r.t. the Lebesgue measure on $\mathbb{R}$.

- Cdfs of integer valued random variables (Binomial, Poisson) are singular
- A monotone $F : \mathbb{R} \to \mathbb{R}$ is differentiable a.e. (m) and $\int_a^b F'(x)dx \leq F(b) - F(a)$

Prop 4.5.1: Given any cdf F, there exist constants $\alpha, \beta, \gamma \geq 0$ with $\alpha + \beta + \gamma = 1$ and cdfs F_d (discrete), F_{ac} (absolutely continuous), F_{sc} (singular continuous) such that

$$F = \alpha F_d + \beta F_{ac} + \gamma F_{sc}.$$

- $\tilde{F}_{ac}(x) \equiv \int_{-\infty}^x F_c'(t)dt$, $\tilde{F}_{sc} \equiv F_c - \tilde{F}_{ac}$, $\tilde{\beta} \equiv \tilde{F}_{ac}(\infty)$; $\beta = (1-\alpha)\tilde{\beta}$, $\gamma = (1-\alpha)(1-\tilde{\beta})$
- α, β, γ are uniquely determined

4.5.2 Cantor ternary set

Def 4.5.2: Let $C_0 = [0, 1]$ and let C_n be the union of the 2^n disjoint closed intervals of length 3^{-n} left after deleting open middle thirds n times. The Cantor ternary set is $C \equiv \bigcap_{n=1}^\infty C_n$.

- $m(C_n) = (2/3)^n$, so $m(C) = 0$ by m.c.f.a.

Theorem 4.5.2: $x \in C$ iff in the ternary expansion $x = \sum_{i=1}^\infty \frac{a_i(x)}{3^i}$ (nonterminating convention), $a_i(x) \in \{0, 2\}$ for all $i \geq 1$.

Corollary 4.5.3: C is in one-to-one correspondence with the set of all sequences of 0's and 1's, and hence with $[0, 1]$.

Remark 4.5.1: C is closed, $m(C) = 0$, uncountable with the cardinality of $[0, 1]$, and nowhere dense.

4.5.3 Cantor ternary function

Define F on $U = C^c$ by $F = \frac{1}{2}$ on $(\frac{1}{3}, \frac{2}{3})$, $F = \frac{1}{4}$ on $(\frac{1}{9}, \frac{2}{9})$, $F = \frac{3}{4}$ on $(\frac{7}{9}, \frac{8}{9})$, and so on; extend continuously to $[0, 1]$.

- F maps $[0, 1]$ onto $[0, 1]$, is continuous and nondecreasing
- $F' \equiv 0$ on U and $m(U^c) = 0$, so F is a singular (continuous) cdf; $\mu_F \perp m$
- With $N(x) = \inf\{i \geq 1 : a_i(x) = 1\}$,

$$F(x) = \sum_{i=1}^{N(x)-1} \frac{a_i(x)}{2^{i+1}} + \frac{1}{2^{N(x)}}$$

- F is the cdf of $X = \sum_{i=1}^\infty \frac{2\delta_i}{3^i}$, δ_i i.i.d. with $P(\delta_1 = 0) = \frac{1}{2} = P(\delta_1 = 1)$; $X \in C$ w.p. 1